SECJ3483 Web Technology
Individual Assignment (15%) & DHL DAC 3.0

Web Application & Robotic Process Automation (RPA)

Academic Session 2025/2026 -2 • Bachelor of Software Engineering with Honours

Important Dates:
Challenge Launch: 31st March 2026
Deadline for Assignment submission: **Friday, 15 May 2026 @5p.m**
Assignment Evaluation Week (FC, UTM): 18 & 21 May 2026
10 Shortlisted Assignment review by DHL: 22 - 26 May
Announcement on the Top 10 shortlisted assignments: 29 May
Presentation (Best 10 Finalist) in Imazium, Petaling Jaya, Selangor: 3 June

1. Introduction - Course Assignment & DHL DAC 3.0 Challenge - Industry Collaboration

This assignment is conducted in collaboration with a real industry partner, DHL Global Finance Services – Asia Pacific, through the DAC 3.0 Challenge.
Students will work on real-world logistics problems inspired by actual industry practices and requirements.
Based on the evaluation results:
• The top 10 projects from UTM JB and 5 projects from MJIIT will be shortlisted for the next stage of the DAC 3.0 Challenge.
• Shortlisted students will represent the faculty in the subsequent phase of the competition.

The overall winners of the DAC 3.0 Challenge will receive:
• Cash prizes
• Certificate of participation
• Internship opportunities at DHL Asia Pacific Shared Services under the Digital Automation Center Team
• Participation in this assignment therefore provides not only academic assessment, but also real industry exposure and potential career opportunities.
• This assignment is part of the Continuous Assessment component for this course and contributes 15% to the overall course grade.

* **Note:** Selection for the DAC 3.0 Challenge is based on industry evaluation and is independent of the academic grading process.

This assignment is weighted:
- 10% Project Management and Progress Tracking
- 50% Web Application Implementation
- 40% RPA Design & Automation Thinking

2. Scenario Allocation by Section

Scenario	Assigned Sections
Scenario 1: AI-Powered Knowledge Base Automation for DHL Logistics Operations	Section 1, Section 2
Scenario 2: AI-Enhanced Incident Reporting & Resolution System	Section 2, Section 4

3. Logistics Scenario Selection

Scenario 1: AI-Powered Knowledge Base Automation for DHL Logistics Operations

DHL Logistics Operations teams handle thousands of messages and documents every day. These come from many different sources, such as:

- Chat messages from MS Teams or Telegram
- Long email threads between teams
- Screenshots used to explain steps or errors
- Handwritten notes or quick instructions shared informally
- Snippets of PowerPoint slides or training materials

All this information is important — but it is unstructured, scattered, and difficult to reuse.
Because of this, creating clean and standardized SOPs (Standard Operating Procedures) or Knowledge Base (KB) articles becomes extremely slow and inconsistent.

Your objective:

You are required to design and build a system that **automates the entire transformation process** — from raw messy input → to clean knowledge articles.

Scenario 2: AI-Enhanced Incident Reporting & Resolution System

DHL's Customer Support teams receive a high volume of incident reports every day—late deliveries, address issues, damaged parcels, system errors, and customer complaints. These reports come through **multiple inconsistent channels**, such as:

- Email inboxes
- Telegram/Teams chat messages
- Phone call notes
- Images/screenshots of damaged packages
- Handwritten instructions from warehouse teams

Because the information is **unstructured, incomplete, and inconsistent**, it is difficult for DHL to:

- Identify the real issue
- Assign incidents to the correct department
- Prioritize incidents correctly
- Track resolution progress
- Produce consolidated reports for management

This leads to slow response times, repeated manual work, and inconsistent customer service quality.

Your objective:

You are required to design and develop an AI-powered Incident Reporting System that automates the full process of collecting, understanding, assigning, and tracking incident reports.

4. System Requirements

4.1 Web Frontend - Web Application (Mandatory)

- Developed using any web technology (example: HTML, CSS, JavaScript, Vue, React, .NET)
- Interactive user interface with event handling
- Use JavaScript functions to manage system logic
- Provide form input and validation
- User-friendly navigation and interface design

4.2 Backend - JSON API OR Database (Mandatory)

Option 1:

- Use JSON format for data storage
- Implement CRUD operations (GET, POST, PUT/PATCH, DELETE)
- All data must be retrieved via API (no hardcoded data)

Option 2:

- All data handled via database storage (example: SQL, MongoDB)

4.3 RPA Component (Mandatory)

- Use UiPath Studio to create an RPA workflow
- Design an RPA workflow diagram
- Explain automation logic clearly

5. Functional Requirements

5.1 Web Application (Mandatory)

- Secured access to the website, allowing RPA to create content
- Upload Console: accepts multiple input types. (minimum: text, PDF and .docx as input)
- Draft and save information with status
- Viewer Page: searchable, filterable (by tag, date, creator, status)
- Versioning: show Draft → Reviewed → Published status with history
- Store creator details: at least basic login for editors/reviewers

5.2 RPA Automation (Mandatory)

- Ingestion: Read new files from Google Drive (or a designated email inbox exported to Drive).
- Duplicate checks: Skip items seen in the last 14 days (hash of text or file).
- Create new content in a web application and be able to attach files/screens
- Update the status in the web application
- Error Handling: Try/Catch + Take Screenshot on failures; write logs.
- Send Summary Email to system admin when executed: totals for created, updated, duplicates, and failed; attach logs.

5.3 Non-Functional Expectations (Optional)

- Integrate the usage of GPT (LLM) to summarize, structure, title, tag, and quality-polish articles; flag conflicts/outdated content; propose step-by-step procedures.
- Draft Builder in Web Application: display AI-proposed Title, Summary, Steps, Tags, Related Links; editors can revise and save.
- Conflict/Outdated information Alerts in Web application: flag if the new draft conflicts with existing published articles.
- Read and transform information from images (PNG/JPG) via GPT as source data for knowledge base update in web application

6. Deliverables and Marks (15%)

A. System/Application Deliverables (8%) : refer Appendix A

- Web Application project source code
- Database or Excel Sheet
- UiPath Project File
- Github project progress

B. Report (8–12 pages) (4%): refer Appendix B

- Introduction and scenario description
- System architecture
- API and JSON structure
- RPA workflow and automation explanation
- User interface screenshots with Description
- Step-by-Step Instructions to run the system (user manual)

C. Demo Video (10-12 minutes) - (3%) : refer Appendix C

- End-to-end system walkthrough with demo of functionality
- Demonstration of CRUD operations via API
- Explanation of RPA automation design

7. Academic Integrity

- Individual work
- Plagiarism is strictly prohibited
- Students may be asked to explain their work

8. Assignment Clone and Commit Instructions

All clones and commits/submissions must be made via GitHub Classroom. The complete repository structure and the link will be provided by the lecturer for each section. Students are required to:

- Accept the GitHub Classroom assignment link provided by the lecturer
- Clone the assigned repository to your local machine
- Develop your project within the provided repository structure
- Commit and push your work regularly throughout the assignment duration
- Ensure all project files are updated in the repository before the deadline

APPENDIX A: MARKING RUBRIC FOR SYSTEM/APPLICATION (8%)

Rubric Scale: 4 = Excellent, 3 = Good, 2 = Satisfactory, 1 = Minimal

A. Web Application & API Implementation (50%)

Criteria	4: Excellent	3: Good	2: Satisfactory	1: Minimal	Weight
A1. System Design & Business Logic	Clear system purpose, well-defined workflow aligned with scenario. Strong justification of design decisions.	Workflow mostly correct with minor gaps or limited explanation.	Basic workflow present but lacks depth or realism..	No clear system logic or poor alignment with scenario.	10%
A2. User Interface & Interaction	Clean, user-friendly UI. Smooth interaction. Responsive updates.	Functional UI with minor usability or interaction issues.	Basic UI with limited interactivity	Poor, static, or unusable interface..	10%
A3. JavaScript Functionality	Well-structured JavaScript logic. Correct use of functions, event handling, and async operations.	Mostly correct logic with minor issues in structure or async handling.	Basic functionality (simple events only, limited logic).	Weak, incorrect, or no meaningful JS logic.	10%
A4. Data Management & Workflow	Proper data structure with full CRUD operations. Includes clear workflow.	Data mostly handled correctly but minor issues in updates or structure.	Basic CRUD only (no workflow or incomplete handling).	Poor data handling or non-functional system.	10%
A5. JSON & API Integration	Full API integration (GET, POST, PUT/PATCH, DELETE). Data dynamically retrieved and updated. No hardcoding.	API mostly works but minor inconsistencies.	Partial API usage (e.g., read-only or limited operations).	No API integration or hardcoded data.	10%

B. RPA Design & Automation (40%)

Criteria	4: Excellent	3: Good	2: Satisfactory	1: Minimal	Weight
B1. RPA Process Identification & Justification	Clearly identifies suitable automation processes with strong justification; aligns perfectly with logistics knowledge-base scenario; shows deep understanding of RPA applicability.	Identifies relevant processes with acceptable justification; shows good understanding.	Basic identification with limited reasoning; justification is shallow or generic	Poor or unclear process identification; justification missing or incorrect	10%
B2. RPA Workflow Design	Detailed, structured, and fully accurate workflow diagram; includes branching, decisions, exception paths, retry logic, and process governance. outputs.	Mostly correct workflow with clear structure; minor gaps in exception or branching logic.	Basic workflow with partial correctness; lacks depth, limited to linear flow.	Minimal or incorrect workflow; very limited understanding of RPA design.	10%
B3. Automation Logic & Trigger Design	Automation logic is well-defined. Demonstrates correct step-by-step actions including ingestion, duplicate checks, web update, error handling, and summaries.	Good logic with clearly described triggers; minor omissions in certain automation steps.	Basic logic explained; trigger mechanism partially defined; lacks details on exception handling.	Minimal logic described; trigger unclear; major missing steps.	10%
B4. RPA Integration & Explanation	Excellent explanation of how RPA integrates with the web application (CRUD actions, API calls, data update, status transition); shows strong understanding of UiPath components.	Good explanation of integration; minor missing details.	Basic integration described; lacks clarity or depth.	Minimal or unclear explanation; fails to show integration understanding.	10%

C. GitHub/Project Progress (10%)

Criteria	4: Excellent	3: Good	2: Satisfactory	1: Minimal	Weight
C1. Commit Progression & Frequency	Clear, consistent commits across development stages (minimum 5), well-distributed over time (not concentrated near deadline), showing logical progression.	Several commits with some progression, but distribution uneven or slightly concentrated near deadline.	Few commits (2–3) with limited progression and weak timeline distribution	Single commit or commits made only near deadline with no clear progression.	5%
C2. Commit Quality & Evidence of Work	Commit messages are clear and meaningful Changes reflect real development progress (not bulk upload).	Some meaningful commits, but mixed with vague messages Evidence of partial structured work.	Mostly vague commit messages with limited indication of structured development.	Poor or no commit messages; work appears uploaded all at once.	5%

APPENDIX B: MARKING RUBRIC FOR REPORT (4%)

Rubric Scale: 4 = Excellent, 3 = Good, 2 = Satisfactory, 1 = Minimal

B1. Report

Criteria	4: Excellent	3: Good	2: Satisfactory	1: Minimal	Weight
B1.Introduction & Scenario Description	Clear and accurate explanation of selected scenario with strong understanding of logistics context.	Mostly clear explanation with minor gaps.	Basic explanation, limited understanding.	Weak or unclear description.	16%
B2. System Architecture & Design	Clear overall architecture (frontend, backend, API, RPA) with well-explained components and data flow.	Mostly clear architecture with minor gaps.	Basic architecture with limited explanation.	Poor or missing architecture.	16%
B3. Technical Explanation (API, Data, JSON)	Clear explanation of API endpoints, JSON structure, and data handling with relevant examples	Mostly complete explanation with minor details missing.	Basic explanation with limited technical depth	Weak or incorrect explanation.	16%
B4. RPA Workflow & Integration Explanation	Clear explanation of RPA workflow, logic, and integration with the system (supported by diagram/screenshots).	Good explanation but lacks some detail.	Basic description with limited clarity.	Weak or unclear explanation.	16%
B5. UI Screenshots & Explanation	Relevant screenshots showing key features/workflow with clear explanation of functionality.	Screenshots provided with limited explanation.	Basic or unclear screenshots.	Missing or poorly explained screenshots	20%
B6. User Manual	Clear, complete setup and usage steps; easy to follow independently.	Mostly clear with minor gaps.	Basic but incomplete; requires assumptions.knowledge to use the system.	Unclear; cannot be used independently.	16%

APPENDIX C: MARKING RUBRIC FOR DEMO/PRESENTATION (3%)

Rubric Scale: 4 = Excellent, 3 = Good, 2 = Satisfactory, 1 = Minimal

C1. Demo/Presentation

Criteria	4: Excellent	3: Good	2: Satisfactory	1: Minimal	Weight
C1. System Demonstration (Functionality)	Complete demo: UI, CRUD, workflow, API, RPA. Smooth and clear.	Mostly complete. Minor issues.	Partial demo. Some features missing.	Very limited or not working.	25%
C2. Explanation & Project Understanding	Clear, confident explanation. Shows strong understanding.	Mostly clear. Minor gaps.	Basic explanation. Limited understanding.	Weak or unclear explanation	25%
C3. RPA Demonstration & Integration	RPA clearly shown. Integration with system is clear.	RPA shown. Integration not fully clear.	Basic or partial RPA demo.	No meaningful RPA demo.	25%
C4. Presentation Quality & Communication	Clear voice. Good pace. Logical flow. Easy to follow.	Generally clear. Minor issues.	Somewhat unclear or disorganized.	Poor or hard to follow.	25%